

■ Powerball Rolling Window Overlap Analysis

Overview

This study analyzes the overlap of white ball numbers across rolling windows of consecutive Powerball drawings. Instead of examining only adjacent drawings, we analyze windows of 2 to 10 consecutive drawings to understand how numbers repeat across different time horizons.

Dataset

Source: Official New York State Powerball Lottery Records

Time Period: January 2010 - July 2026

Total Drawings: 1,967

Data Quality: No duplicates found, fully verified and cleaned

Methodology

For each window size (2 through 10 consecutive drawings):

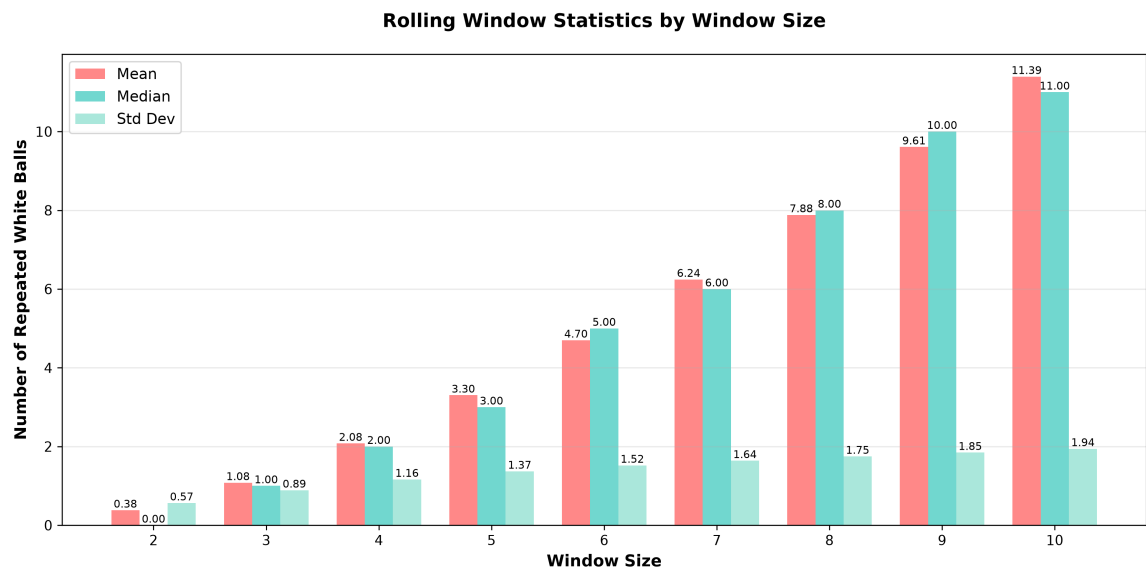
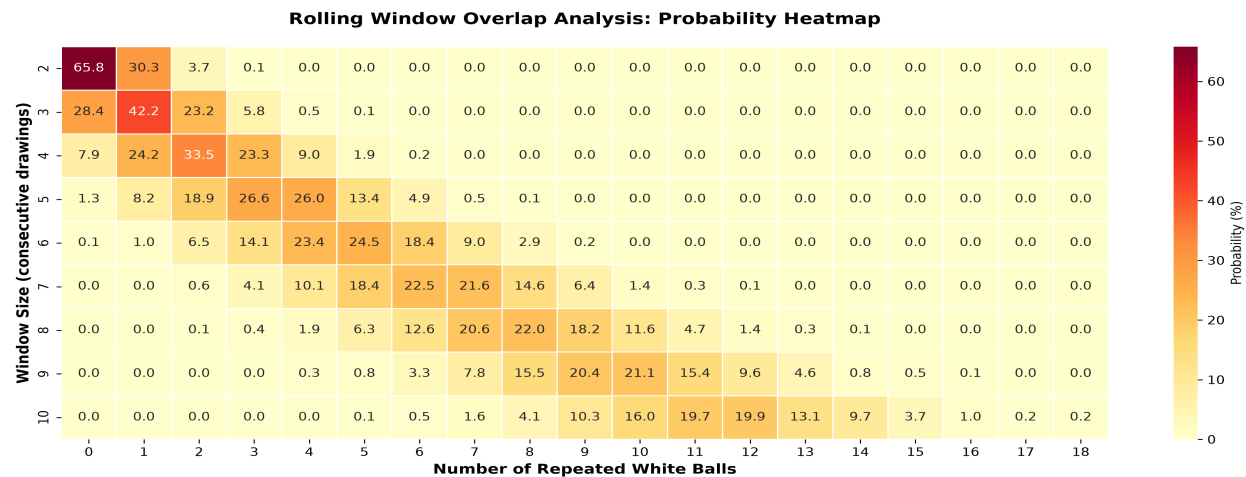
- Extract all white ball numbers from each drawing in the window
- Count how many numbers appear more than once across the window
- Calculate statistics: mean, median, min, max, standard deviation
- Build probability distribution for each window size
- Construct empirical probability matrix

Results

Probability Matrix (Empirical)

	Repeated_F1	Repeated_F2	Repeated_F3	Repeated_F4	Repeated_F5	Repeated_F6	Repeated_F7	Repeated_F8	Repeated_F9	Repeated_F10	Repeated_F11	Repeated_F12	Repeated_F13	Repeated_F14
	3.71	0.15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	23.16	5.75	0.51	0.05	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	33.5	23.27	8.96	1.88	0.25	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	18.95	26.64	25.98	13.4	4.89	0.51	0.05	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	6.52	14.07	23.39	24.52	18.35	8.97	2.85	0.2	0.0	0.0	0.0	0.0	0.0	0.0
	0.61	4.08	10.05	18.36	22.54	21.62	14.58	6.37	1.43	0.31	0.05	0.0	0.0	0.0
	0.05	0.36	1.94	6.28	12.55	20.61	22.04	18.16	11.58	4.69	1.38	0.31	0.05	0.0
	0.0	0.0	0.31	0.77	3.27	7.76	15.52	20.42	21.08	15.36	9.6	4.59	0.82	0.46
	0.0	0.0	0.0	0.05	0.46	1.58	4.14	10.32	15.99	19.71	19.87	13.13	9.7	3.68

Visualizations



Rolling Window Analysis: Summary Statistics

Window Size	Mean Overlaps	Median	Min	Max	Std Dev
2	0.382	0.0	0	3	0.565
3	1.080	1.0	0	5	0.890
4	2.078	2.0	0	6	1.164
5	3.303	3.0	0	8	1.370
6	4.700	5.0	0	9	1.521
7	6.243	6.0	2	12	1.639
8	7.884	8.0	2	14	1.749
9	9.607	10.0	4	16	1.846
10	11.395	11.0	5	18	1.944

Interpretation

Key Observations:

- As window size increases, the expected number of repeated white balls increases
- Window size 2: Average 0.38 repeated numbers
- Window size 10: Average 11.39 repeated numbers
- The probability distribution shows clear patterns:
 - Smaller windows concentrate around lower overlap counts
 - Larger windows show broader distributions
- This is consistent with probability theory: longer windows have higher likelihood of repeated numbers

Limitations

- This analysis describes historical empirical distributions only
- No predictive claims about future drawings
- Results are specific to the time period analyzed (2010-2026)
- Drawing mechanism may evolve over time
- Lottery outcomes are inherently random and unpredictable